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Assessment of light adequacy for vertical farming in a tropical city

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ABSTRACT

The pursuit of urban agriculture as part of a city's green infrastructure is often a challenge, particularly within compact cities, where there is a limited amount of space between buildings for urban farming and gardening. Instead, such high-rise urban developments present often under-utilized spaces on the vertical surfaces of buildings. A key unknown is the adequacy of light for plant growth. Many leafy vegetables that require high amounts of light form a significant proportion of the staple diet in many Asian countries. We report on the assessment of sunlight adequacy for growing leafy vegetables in a compact tropical city, based on the high-rise and high-density residential environment of Singapore. Leaf physiological traits of seven leafy vegetables were assessed and used to estimate plant light requirements. A survey of photosynthetically active radiation (PAR) along exposed corridors showed that the daily light integral (DLI) value ranged from 2 to 35 mol m^{-2} d^{-1} under relatively ideal weather conditions during days with abundant solar insolation, and façades that experienced a minimum of half-day direct insolation matched the light requirements of vegetables within the moderate to very high-light categories. With regard to the building form, PAR increased gradually with height, but remains highly influenced by façade orientation and configuration. Owing to the annual north–south oscillation of the sun's path, reduced annual PAR variability and higher total annual PAR at façades, buildings with an east–west orientation will better support continuous vegetable cultivation, especially for basic building typologies without self-shading configurations. However, excessive PAR and temperatures during mid-day hours may hinder plant growth. By highlighting such patterns in levels of PAR, this study confirms the potential for high-rise and high-density conditions in the tropics to support farming using typically under-utilized vertical spaces of residential buildings.

1. Introduction

The rise of urban agriculture as an important form of green infrastructure has accompanied global efforts to increase the sustainability and resilience of cities in recent years (Lin et al., 2017). Indeed, negative pressures on food supply have led to the practice of urban farming as a possible strategy to improve food security by complementing rural agriculture in supplying food to urban dwellers. This typically includes ground-level farming or gardening in private, rooftop, or community gardens. However, these are often challenging to undertake in compact cities, where there are limited amounts of interstitial green spaces between buildings and competing demands on rooftop spaces for building utilities. Given these constraints, under-utilized vertical spaces along building façades have been explored for the implementation of vertical farming such as conventional low-cost micro-gardening, building-integrated edible landscaping, as well as vertical greenhouse systems (Kalantari et al., 2017).

In spite of the technological feasibility and desirability of vertical farming, one key unknown is adequacy of light for plant growth; in particular, photosynthetically active radiation (PAR) between 400 and 700 nm of the solar spectrum (Ross and Sulev, 2000). Studies have shown that the distribution of instantaneous PAR (measured as photosynthetic photon flux density or PPFD, in μmol m^{-2} s^{-1}) differs significantly from sunshine-hour duration (Mercado et al., 2009; Tan et al., 2015a). Hence, knowledge of PAR conditions in the environment can help us better understand the potential for growing plants that require high amounts of light, such as leafy vegetables (Faust, 2002; He, 2015).

Abbreviations: N, north-facing façade; *S*, south-facing façade; *EN*, east-facing façade with northward obstruction; *ES*, east-facing façade with southward obstruction; *WN*, west-facing façade with northward obstruction; *WS*, west-facing façade with southward obstruction.

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The light requirements for ornamental plants, turf grasses, and several plant crops are known (Torres and Lopez, 2010; Warrington and Norton, 1991; Wolff and Coltman, 1990), often expressed as the cumulative daily PAR (mol m^{-2} d^{-1}) over a 24-h photoperiod, or the "Daily Light Integral" (DLI) in production horticulture. However, there is little information for the vegetables that form a significant proportion of the staple diet in many Asian countries (Ooraikul et al., 2008).

With a high abundance of sunlight in the tropics, studies have assessed PAR conditions at interstitial building spaces in dense urban environments (Tan et al., 2015a, 2014), and explored the suitability of rooftop spaces for the growth of leafy vegetables (Astee and Kishnani, 2010; He, 2015). However, limited research has investigated the suitability of the light environment along vertical spaces, which are highly affected by building form and spatial factors that influence shading patterns and the availability of direct and diffused incident radiation (Hii et al., 2011; Valladares-Rendón and Lo, 2014). PAR is also subject to both diurnal and annual temporal variations which depend on factors such as time of day, weather, as well as seasonal variability caused by the annual solar cycle. The ability of vertical spaces to support the growth of vegetables is thus an unknown and a possible limitation to urban farming, even under high light conditions in the tropics.

The aim of this study was to evaluate the spatio-temporal characteristics of PAR for growing leafy vegetables at high-rise vertical spaces, using tropical Singapore as a case-study location. Amongst compact cities across the world, Singapore is an example of a city-state that lacks a domestic hinterland for agricultural self-sufficiency through conventional farming. A significant portion of unused vertical space in Singapore includes exposed corridors, balconies, and windows along the building façades at high-density, high-rise public housing residential estates. There are thus opportunities to explore the suitability of PAR conditions along these vertical spaces, and the suitability of such environments for vertical farming. The specific objectives of the study were: (1) Investigate the effect of building form (height, configuration, and façade orientation) on PAR, along exposed accessible corridors of public housing apartment block façades; (2) Estimate the DLI requirements of selected vegetable crops based on leaf physiological traits, and (3) Assess the suitability of such spaces in providing adequate PAR for optimal growth of vegetables.

2. Materials and methods

2.1. Leaf-level gas exchange analysis

Table 1 shows varieties of fast-growing leafy vegetables grown and sold in Singapore and Southeast Asia (Herklots, 1972; Jansen et al., 1993; Kok et al., 1991), which were grown at the National University of Singapore Native Plant Nursery (1°17′43.9"N 103°46′28.4"E) over the course of six weeks from September–October 2015. Local seed varieties from Ban Lee Huat Seed Pte Ltd were used. Black netting was used to reduce excessive radiation to approximately 75% of unshaded conditions. The DLI value within the nursery typically ranged from 5 to 25 mol m^{-2} d^{-1}, with an average of 12–14 mol m^{-2} d^{-1}.

Table 1
List of vegetable crops selected for analysis.

Vernacular Name	*Botanical Name*
Bayam (Chinese spinach or red amaranth)	*Amaranthus tricolor* L.
Cai xin (Chinese flowering cabbage)	*Brassica rapa* L. subsp. *chinensis* (L.) Hanelt var. *parachinensis* (L. H. Bailey) Hanelt
Chinese cabbage (Napa cabbage)	*Brassica rapa* L. subsp. *pekinensis* (Lour.) Hanelt
Kailan (Chinese kale)	*Brassica oleracea* L. var. *alboglabra* (L. H. Bailey) Musil
Kang kong (Water spinach/convolvulus)	*Ipomoea aquatica* Forssk.
Lettuce	*Latuca sativa* L.
Pak choy (Chinese chard)	*Brassica rapa* L. subsp. *chinensis* (L.) Hanelt

Leaf photosynthetic responses were measured using an open-flow CO_2 gas exchange system (LI 6400, Li-Cor Inc., Lincoln, Nebraska, USA) fitted with the standard leaf chamber, CO_2 injector system and red/blue LED light source (6400–02 B LED). Measurements were made upon the maturation of each crop type, within the edible phase of the growing period. Measurements were taken for the youngest mature leaves of three different individuals per crop type from 0800 − 1700 h. To derive the light response curve, stepwise changes made in PPFD were made in the following order: 250, 100, 50, 25, 250, 500, 750, 1000, 1500, 2000, 2500 μmol m^{-2} s^{-1}. In order to create stable measurement conditions, measurements were made under a block temperature of 28 °C, constant CO_2 level of 360 μmol mol^{-1}, and an air-flow rate adjusted between 300 and 500 μmol s^{-1} to reduce the effect of humidity on measurement error. Leaves were allowed to acclimatise for 2–5 min at each PPFD level, and measurements were logged after readings had stabilized. Using the non-linear regression function in R 3.2.3 (R Core Team, 2015), measurements obtained for each individual were fitted to a non-rectangular hyperbola formula described in Baltzer and Thomas (2007) to derive a modelled light response curve. R^2 larger than 0.97 was achieved for all the fitted light response curves for each individual measured. From the fitted functions, the key variables of the light response curve, namely the light saturation point (LSP) and light compensation point (LRC) were estimated alongside other physiological variables. Duncan's new multiple range test was used to test for significant differences between crops.

Commercial greenhouse production utilizes supplementary lighting to achieve constant PAR levels that match plant's DLI requirements (Torres and Lopez, 2010). Based on this approach, the DLI requirements for each crop were estimated based on the equations:

$$\textit{Target DLI} = \textit{mean LSP at 70\% of maximum photosynthesis} \times 0.0432$$

$$\textit{Maximum DLI} = \textit{mean LSP at 85\% of maximum photosynthesis} \times 0.0432$$

where the target DLI value is the level of cumulative PAR over a 12-h photoperiod required for daily growth. It was calculated based on the LSP at 70% of maximum photosynthesis, as there were insufficient differences between crop LSP values at lower percentages. The maximum DLI value is the maximum level of cumulative PAR over a 12-h photoperiod for daily growth. It was calculated based on the LSP at 85% of maximum photosynthesis, as higher percentages resulted in unreasonably high PFFD values (i.e., 3000–7000 μmol m^{-2} s^{-1}). The value 0.0432 represents the conversion factor for PAR (μmol m^{-2} s^{-1}) to DLI value (mol m^{-2} d^{-1}) over a 12-h photoperiod.

These equations assume an ideal, controlled environment for factors such as temperature and humidity, that the crops grow optimally at constant PAR levels throughout a 12-h photoperiod, and that fluctuations in the PFFD are minor and do not exceed levels that negatively affect the crops. Unlike woody plants, leafy vegetables are relatively small, less morphologically complex, and do not invest significant amounts of photosynthates for structural growth and the formation of woody tissues (Chapin et al., 1990). Also, increased light requirements with plant age owing to self-shading and phenology is less pronounced in vegetable crops, because of their relatively short edible growth phase and life span (Gerrish, 1990; Thomas and Wien, 1997). Thus, it was assumed that the leaf-level LSP is a reasonable estimate of the whole-plant light saturation point (Baltzer and Thomas, 2007).

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2.2. PAR measurement and analysis

This study examines the effect of building form (i.e. height, configuration, and façade orientation) on PAR. Amongst the diverse public housing building typologies in Singapore, basic configurations such as Linear, Point Block and L-shape block arrangements built before the 1990s generally provide more predictable conditions, because of minimal self-shading building geometry and more spacious, exposed corridors suitable for micro-gardening. Apartment blocks 861, 861A, and 859 along Tampines Avenue 5 (1°21′17.5“N 103°56′15.5“E) with well-defined façade orientations were selected as study sites (Fig. 1). Like most public housing apartment blocks in Singapore, the apartment blocks surveyed are oriented to allow a larger surface area to face the north and south direction (Housing and Development Board, 2014). Owing to the presence of self-shading building geometry for the L-shaped configuration, east- and west-facing façades with both north- and southward obstructions were surveyed. These well-defined parameters may serve as a general guide to extrapolate PAR conditions along façades of other building configurations.

Instantaneous PAR at corridors along building façades were measured using quantum sensors (Onset Photosynthetic Light (PAR) Smart Sensor S-LIA-M003; Bourne, MA, USA), with a sampling interval of 10 s and the mean values logged at 1-min intervals (Onset HOBO Micro Station Data Logger H21-002; Bourne, MA, USA). Upward-facing sensors were placed directly above the parapet on the outer edge of the railing of exposed corridors, and were aligned vertically along each building façade. This also served as an approximation of PAR at the corridor area immediately next to the parapet wall, which is usually used for corridor gardening. Sensors were placed at heights of 4.6–35.4 m above ground level.

Building façades were surveyed in pairs, each for a duration of two weeks–north- and south-facing façades (*N* and *S*) were surveyed from 12–24 October 2015; east-facing façades with north and southward obstruction (*EN* and *ES*) were surveyed from 26 October–7 November 2015; west-facing façades with north and southward obstruction (*WN* and *WS*) were surveyed from 9–25 November 2015. All storeys within each building façade were surveyed during the same time period. In order to control for weather variability across the different survey periods, one day was selected per period based on comparability in weather conditions: low cloud cover, rainfall and pollutant standards index (PSI) levels, generally smooth minute-by-minute instantaneous PAR levels, as well as a high total cumulative daily PAR levels were ensured. 15 October (façades *N* and *S*), 3 November (façades *EN* and *ES*) and 22 November (façades *WN* and *WS*) were selected for comparison across building façades surveyed at different periods. Conditions surveyed thus represent PAR availability under relatively ideal weather conditions during days with abundant solar insolation. As for comparisons between the daily maximum PAR levels, the Kruskal-Wallis one-way analysis of variance was used to test the differences in means across the entire survey period, between storeys within each façade; post-hoc pairwise comparisons were conducted using the Nemenyi test. The Brown-Forsythe Levene-type test was used to test for variance homogeneity. All statistical analyses were performed in R 3.2.3 (R Core Team, 2015).

3. Results

3.1. Estimated DLI requirements

The best-fit light response curves for each crop are shown in Fig. 2. Despite variation between individuals for each crop, an R^2 value larger

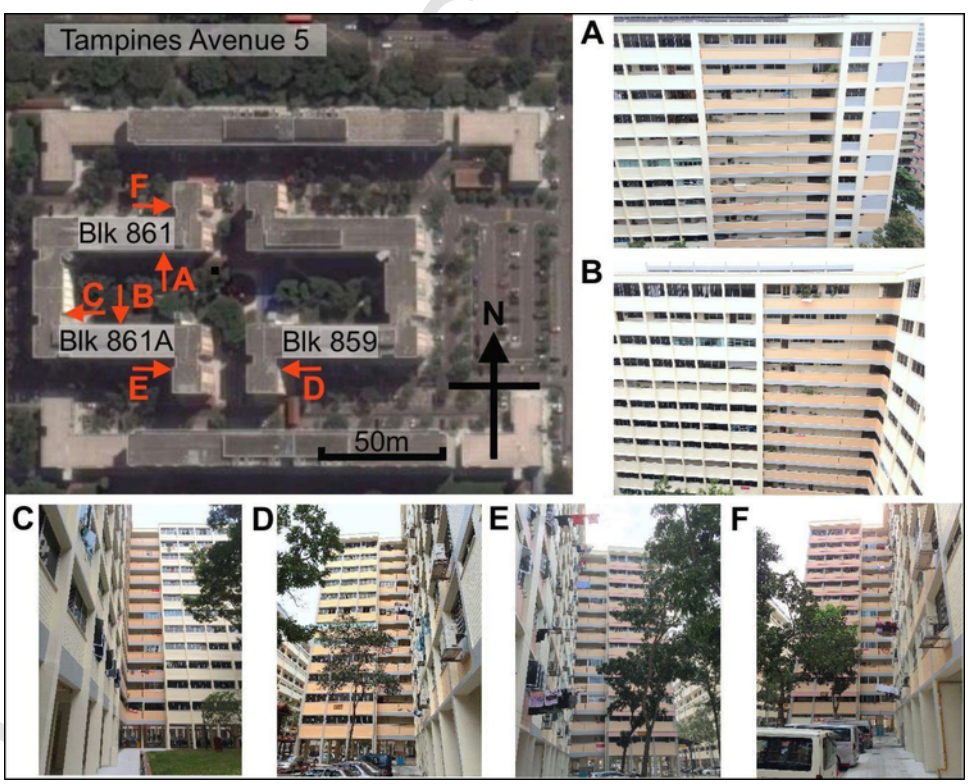

Fig. 1. Aerial view of study site and images of building façades surveyed. **A** South-facing façade of Block 861 (*S*); **B** North-facing façade of Block 861A (*N*); **C** East-facing façade of Block 861A with southward obstruction (*ES*); **D** East-facing façade of Block 859 with northward obstruction (*EN*); **E** West-facing façade of Block 861A with northward obstruction (*WN*); **F** West-facing façade of Block 861 with southward obstruction (*WS*). Arrows indicate the direction from which the photographs were taken. The horizontal distance between façade *N* and *S* is 30 m. The aerial view of the study site was adapted from Google, DigitalGlobe (2015). *(2-column-fitting image, web-version in color).*

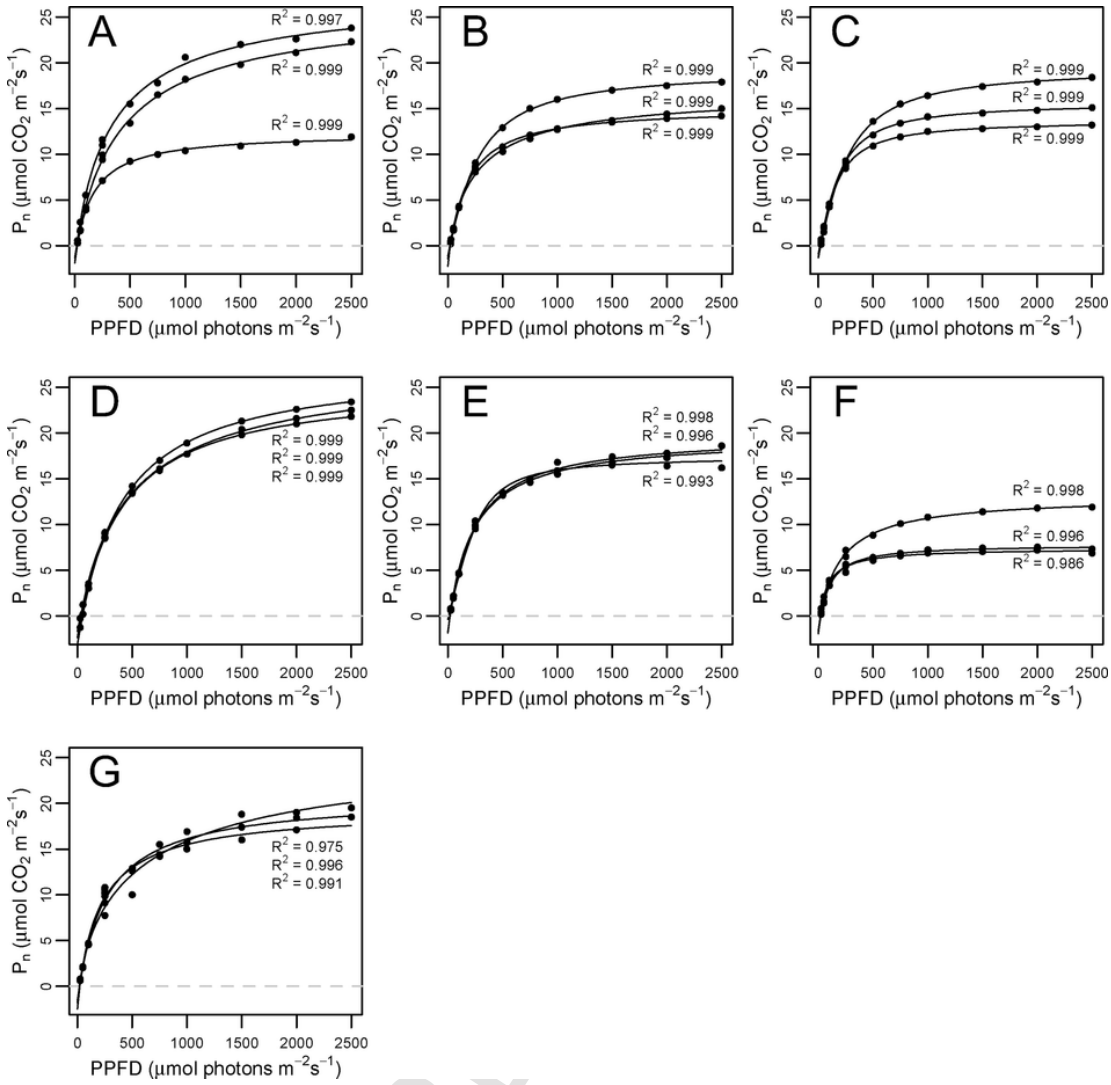

Fig. 2. Net photosynthetic rate (P_n) against photon flux density (PPFD) for three individuals per crop type. **A** Bayam; **B** Cai xin; **C** Chinese cabbage; **D** Kailan; **E** Kang kong; **F** Lettuce; **G** Pak choy. Data points denote the photosynthesis level measured at each irradiance level, while the curves are best fits to the nonrectangular hyperbolic equation in Baltzer and Thomas (2007). *(1.5-column-fitting image).*

than 0.97 was achieved across the fitted functions. The LSP and photosynthetic capacity (A_{max}) were lowest in lettuce and highest in kalian (more information on each physiological variable can be found in the Supplementary Information). Referring to Table 2, the target and maximum DLI values derived from the mean LSP values ranged from 14.51 mol m^{-2} d^{-1} (lettuce) to 47.22 mol m^{-2} d^{-1} (kailan), and 34.34 mol m^{-2} d^{-1} (lettuce) to 113.51 mol m^{-2} d^{-1} (kailan), respectively. Plant DLI requirements are typically categorised into very low light (less than 5 mol m^{-2} d^{-1}), low light (5–10 mol m^{-2} d^{-1}), moderate light (10–20 mol m^{-2} d^{-1}), high light (20–30 mol m^{-2} d^{-1}), and very high light (30–60 mol m^{-2} d^{-1}) (Faust, 2002; Torres and Lopez, 2010). Based on estimations, the categorisation of the target DLI requirements for the leafy vegetables are as follows: kailan, pak choy, and bayam grow well under very high amounts of light being 47.22, 39.96, and 33.95 mol m^{-2} d^{-1}, respectively; cai xin grow well under high amounts of light at 24.51 mol m^{-2} d^{-1}; and kang kong, Chinese cabbage, and lettuce grow well under moderate amounts of light, being 19.90, 17.35, and 14.51 mol m^{-2} d^{-1}, respectively.

Table 2
Target and maximum daily light integral (DLI) values for each crop type. The mean ± one standard error for three individuals are shown (mol m^{-2} d^{-1}). The light saturation point (LSP) used in the calculation of target and maximum DLI value was the irradiance level at 70% and 85% of maximum photosynthesis respectively. Superscript letters denote statistically significant differences between groups of crops, based on critical values derived using Duncan's new multiple range test. The target DLI value was used to assign each crop to DLI categories in Faust (2002).

Crop type	Target DLI value	Maximum DLI value	DLI category
Bayam [abc]	33.95 ± 8.43	83.74 ± 22.11	Very high light
Cai xin [abc]	24.51 ± 3.33	57.56 ± 10.47	High light
Chinese cabbage [c]	17.35 ± 2.57	35.71 ± 5.51	Moderate light
Kailan [a]	47.22 ± 3.48	113.5 ± 11.18	Very high light
Kang kong [bc]	19.90 ± 4.37	47.60 ± 9.91	Moderate light
Lettuce [c]	14.51 ± 4.16	34.34 ± 10.51	Moderate light
Pak choy [ab]	39.96 ± 13.15	102.9 ± 37.88	Very high light

3.2. Effect of building height and façade orientation on PAR

Fig. 3 shows the DLI value against building height along each building façade, based on PAR values obtained on the day selected for comparison across the three survey periods. Under relatively ideal weather conditions during days with abundant solar insolation, the DLI value ranged from 2 to 35 mol m^{-2} d^{-1}. With the exception of façade *S*, the DLI value at building façades surveyed were mostly below the average unshaded daily PAR (29.3 mol m^{-2} d^{-1}) in Singapore, which was measured on a fully unshaded roof (estimated 33 m above sea level, 1°17′57.6“N 103°46′16.1“E) between 17 October 2012–23 December 2013 (Tan et al., 2015a). Façades unobstructed from the southward direction (*S, EN, WN*, shown as black line plots) had consistently higher DLI value readings compared to south-obstructed façades (*N, ES, WS,* shown as red line plots). Insolation was observed to originate from the southward direction, and a lack of direct insolation was observed along south-obstructed façades. Analysis of the minute-by-minute diurnal instantaneous PAR distribution (Fig. 4) shows that variability in PAR was much higher at façades unobstructed from the southward direction. Owing to building shade, sharp declines and rises in instantaneous PAR levels during mid-day hours were observed along east- and west-facing façades *EN* and *WN*, respectively. In spite of this, east- and west-facing façades obstructed from similar directions (*EN* and *WN*; *ES* and *WS*) had largely similar DLI values (Fig. 3).

The DLI value generally increased across building height (Fig. 3), with the exception of the low DLI value measured at the top storeys of south-unobstructed façades (*S, EN, WN*) where overhanging sunshade reduced PAR. Indeed, storey 13 received lower levels of instantaneous PAR (Fig. 4, blue lines) along these façades. Conversely, the lack of an overhanging sun-shade at façade *ES* allowed the top storey to receive direct insolation up till mid-day hours, resulting in a considerably higher DLI value. Finally, shade cast by trees was observed to be a contributing factor to the DLI trend reversal at lower heights along exposed corridors in façade *WS* (Fig. 3).

In matching crop DLI requirements, a comparison with the mean maximum DLI estimates for each crop (Table 2) revealed that the DLI values along intermediate building heights of the façade *S* slightly exceeded the maximum DLI value for lettuce. The building façades generally fell within the following DLI categories: Very high-light at the south-facing façade (*S*); moderate to high-light at east- and west-facing façades unobstructed from the southward direction (*EN, WN*); very low to low-light at south-obstructed façades (*N, ES, WS*).

Comparing between the instantaneous PAR distribution and the LSP of crops (Fig. 4), PAR levels exceeded the LSP of lettuce, Chinese cabbage, kang kong, and cai xin, particularly during mid-day hours at façades unobstructed from the southward direction (*S, EN, WN*), as well as the top storey of façade *ES* (Fig. 4(d)). PAR levels did not exceed the LSP of bayam, pak choy, and kailan. Analysis of the daily maximum instantaneous PAR at each location averaged across the survey period showed a similar trend (Fig. 5).

Finally, it was noted that the presence of haze[1] during the survey period resulted in highly positive correlations between the daylight-hour Pollutant Standards Index (PSI) and DLI values at locations that did not experience direct insolation. These correlations were largely unaffected for façades that experienced direct insolation. These observations are consistent with the existing literature, where diffused PAR irradiance has been shown to increase 7–51% under turbid atmospheric conditions (air pollution) caused by urban aerosols (Jacovides et al., 1997). Thus, increased diffuse radiation on days with high PSI could have contributed to the relatively higher instantaneous PAR and DLI values along corridors that lacked direct insolation.

[1] Singapore was affected by the Southeast Asian haze crisis from June–October 2015.

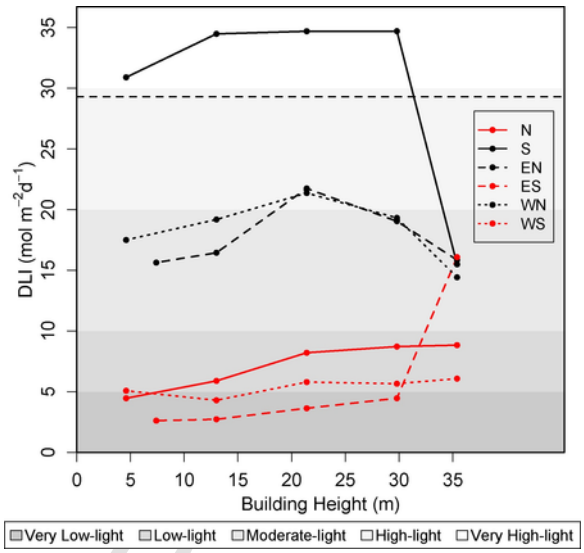

Fig. 3. Daily light integral (DLI) values against building height along building façades ***N*** (north-facing); ***S*** (south-facing); ***EN*** (east-facing with northward obstruction); ***ES*** (east-facing with southward obstruction); ***WN*** (west-facing with northward obstruction); and ***WS*** (west-facing with southward obstruction). DLI values represent relatively ideal weather conditions during days with high solar insolation. Building façades are categorised into one of two generic configurations—unobstructed from the northward direction (red lines), and unobstructed from the southward direction (black lines). Individual data points for each storey surveyed and their corresponding building heights are shown (storeys 2, 5, 8, 11 and 13). Shaded zones represent DLI categories in Faust (2002). The horizontal black dashed line shows the average unshaded cumulative daily PAR (29.3 mol m^{-2} d^{-1}) in Singapore (Tan et al., 2015a). *(single-column-fitting image, in color).* (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

4. Discussion

The cumulative daily PAR or DLI value at a location generally ranges from 5 to 50 mol m^{-2} d^{-1}, and may reach up to 60 mol m^{-2} d^{-1} during summer in temperate zones (Korczynski et al., 2002), owing to a prolonged photoperiod. Within greenhouses, reduced light transmission to 35–70% of full sunlight typically results in a DLI value range of 1–35 mol m^{-2} d^{-1} (Dorais, 2003). This was very similar to the DLI value along the building façades surveyed under relatively ideal weather conditions (2–35 mol m^{-2} d^{-1}), which were generally less than the average unshaded cumulative daily PAR (29.3 mol m^{-2} d^{-1}) in Singapore (Tan et al., 2015a).

4.1. Crop light requirements

Studies have investigated the suitability of the DLI conditions for growth of various ornamental plants (Torres and Lopez, 2010) and food crops (Table 3). Owing to the paucity of literature on the DLI requirements of local leafy vegetables, these were estimated by calculating irradiance at varying levels of maximum photosynthesis (Table 2). As LSP levels at high percentages of maximum photosynthesis showed extreme variation to values well beyond typical DLI levels, the target DLI value based on LSP values at 70% of maximum photosynthesis were used to estimate the relative light requirements between the vegetables selected, and to compare to the DLI within the built environment. The target DLI values correspond the high light levels above 12 mol m^{-2} d^{-1} required for the cultivation of many vegetables (Faust, 2002; He, 2015; see Table 3).

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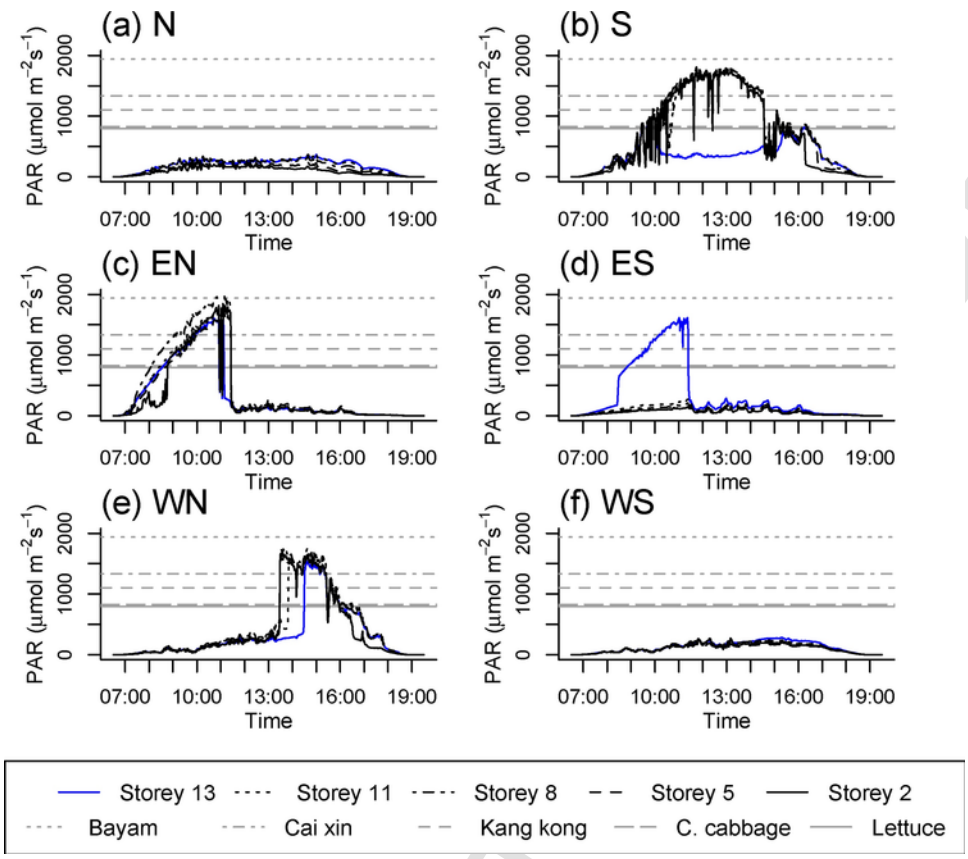

Fig. 4. Diurnal instantaneous PAR distribution along building façades **(a)** ***N*** (north-facing); **(b)** ***S*** (south-facing); **(c)** ***EN*** (east-facing with northward obstruction); **(d)** ***ES*** (east-facing with southward obstruction); **(e)** ***WN*** (west-facing with northward obstruction); and **(f)** ***WS*** (west-facing with southward obstruction). PAR levels at storeys 2, 5, 8, 11 and 13 are shown for each façade, under relatively ideal weather conditions during days with high solar insolation. Horizontal grey lines show the mean leaf-level light saturation point (LSP) of crops within the PAR measurement range, calculated at 85% of maximum photosynthesis. *(2-column-fitting image, in color).*

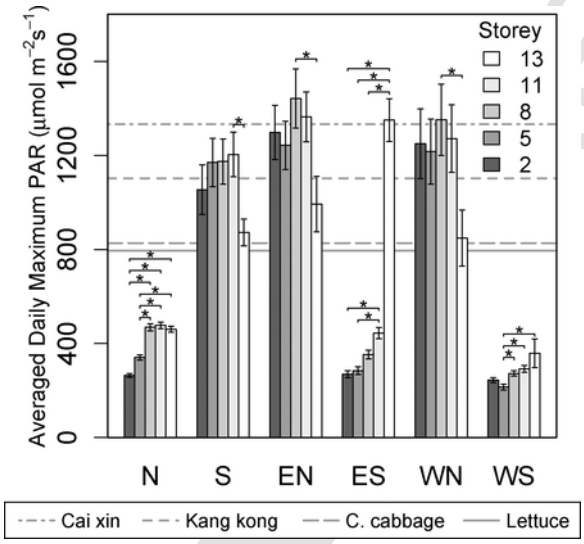

Fig. 5. Average daily maximum instantaneous PAR across the survey period for each storey at façades (a) *N* (north-facing); (b) *S* (south-facing); (c) *EN* (east-facing with northward obstruction); (d) *ES* (east-facing with southward obstruction); (e) *WN* (west-facing with northward obstruction); and (f) *WS* (west-facing with southward obstruction). PAR at east-facing façades was surveyed at storey 3 instead of storey 2. Error bars show ± one standard error. Horizontal grey lines show the mean leaf-level light saturation point (LSP) of crops within the PAR measurement range. Asterisks (*) indicate significant differences ($P < 0.05$) for pairwise comparison between storeys. *(single-column-fitting image).*

Table 3
Daily light integral (DLI) requirements of selected food crops. DLI data were adapted from Dorais (2003) and Chang et al. (2008).

Plant Crop	Botanical Name	Suitable DLI Value (mol m^{-2} d^{-1})	DLI Category
Cucumber	*Cucumis sativus* L.	5.5–30	Low–high light
Lettuce	*Latuca sativa* L.	≥ 12	Moderate light
Sweet basil	*Ocimum basilicum* L.	10–30	Moderate–high light
Sweet pepper	*Capsicum annum* L.	≥ 12	Moderate light
Tomato	*Solanum lycopersicum* L.	≥ 30	Very high light

Despite similarities between estimated DLI requirements and values based on actual growth experiments, there exist limitations to the results obtained. The estimated DLI requirements do not reveal information about the photoperiod, which may affect plant growth (Hillman, 1956; Thomas and Vince-Prue, 1997). For example, higher light intensities can reduce plant growth cycles, as shown in a reduction of the growing period for cucumber from 24 to 10 days with an increase of DLI from 5.5 to 30 mol m^{-2} d^{-1} (Dorais, 2003). Additionally, the use of supplementary lighting to increase the photoperiod (and hence, DLI) to 16–24 h and to spread out the intensity of instantaneous PAR levels enabled increased biomass in numerous crops (Demers et al., 1998; Dorais et al., 1996). However, it should be noted that negative symptoms such as leaf chlorosis may develop with excessively long photope-

riods (Dorais et al., 1995). It was thus assumed that the estimated DLI requirements represented a 12-h photoperiod for comparison with natural light conditions in tropical and equatorial Singapore (Decoteau, 1998; Yoshida, 1973).

In addition to plant DLI requirements, information about a plant's ability to acclimatise to varying irradiance levels is an important consideration for crop cultivation. This includes the ability of crops to adapt to varying levels of cumulative PAR, as well as abrupt changes to instantaneous PAR. Poor light acclimation in certain plants may result in low-light stress or photoinhibition under insufficient or excessive levels of PAR, respectively (Björkman, 1981). In this regard, LSP values can provide information about the ability for plants to withstand high irradiation levels. Indeed, high light and temperatures have shown to compromise lettuce quality and productivity owing to tipburn (Frantz et al., 2004), and the relatively low LSP values observed for lettuce suggest that it is unable to withstand high PAR levels. The reverse is true for crops with higher LSP values such as bayam, pak choy, and kalian. Notably, bayam as a C_4 plant would likely be better able to survive hot and dry conditions, owing to its ability to photosynthesize after leaf stomata have closed for moisture conservation (Lin and Ehleringer, 1983). However, while the LSP value may allow us to infer the ability of crops to acclimatise to high levels of instantaneous PAR, further growth experiments under controlled environmental conditions are needed. In particular, a better understanding of crop adaptation to abrupt changes in light levels is needed, given the rapidly changing light conditions in the urban environment caused by building shade. For example, even though lettuce requires lower DLI conditions (Wolff and Coltman, 1990), its poor acclimation to low-light levels may result in poor growth under changing light conditions on cloudy days (Frantz and Bugbee, 2005).

Finally, it should be noted that plant light requirements are affected by factors such as water availability, temperature, and levels of carbon dioxide in the environment (Christiaens et al., 2014; Demmig-Adams and Adams, 2006; Frantz et al., 2004). Faust (2002) described the cessation of transpiration and photosynthesis due to closure of plant stomata in plants under drought stress, as well as the influence of temperature on the speed of plant development. Even though greenhouse vegetables require high-light levels, excessive temperatures may limit yield and contribute to heat stress and photoinhibition. As minimum light levels in plants are lower at cool temperatures, high-light and cool temperatures enable the most energy to be channelled toward growth, resulting in large, high-quality plants (Faust, 2002). In a study by He and Lee (1998) under tropical conditions, plants with rootzones that were cooled to lower temperatures showed higher levels of maximum photosynthesis rates. Furthermore, factors such as temperature also affect bolting, which can negatively influence the size, texture and taste of leaves (He and Lee, 1998). While there have been studies that simulate light, temperature and water regimes in temperate regions (Moccaldi and Runkle, 2007; Pramuk and Runkle, 2005), average daily temperatures within greenhouses in Singapore range from 25 to 36 °C and can reach up to 42 °C on bright sunny days (He, 2015). In view of this, more can be done to improve our understanding of optimal growth conditions for comparison with environmental conditions unique to the tropical urban environment (Song et al., 2017).

4.2. PAR conditions along building façades

Basic building typology and well-defined façade orientations surveyed in this study provide a basis for general predictions of PAR levels along vertical spaces within the built environment. The basic linear and L-shaped building configurations of older public housing apartment block morphologies that were selected are distinctly different from the narrower, sheltered corridors in newer apartment blocks, which are also generally less suitable for micro-gardening. In addition to PAR availability under ideal weather conditions, taking into account annual variations in PAR caused by the sun's north–south oscillation, cloud cover, and rainfall over a longer survey period would have been preferred. Nevertheless, according to Tan et al., (2015b), seasonal variation in PAR between months was shown to be minimal when compared to the annual average, given the equatorial position of Singapore. Thus, the effects of shading would likely outweigh the effects of seasonal variation in PAR.

Based on the results, the influence of façade orientation and building forms on PAR is considerably higher compared to building height. The slight increase in the DLI with building height can likely be attributed to diffused incident radiation from the sky, as well as reduced shading from neighbouring buildings (Hii et al., 2011). However, PAR remains highly affected by the built environment, as sources of shade such as overhanging sunshade and nearby trees resulted in drastic reductions in the DLI value. Factors such as a building's external, internal and long-axis orientation have shown to exert a strong influence on PAR (Tan et al., 2015a; Zhang et al., 2012), while reduced spacing between buildings has been associated with increased shading and reflected insolation (Hii et al., 2011). Notably, across numerous building typologies, the basic point block configuration was found to result in the highest insolation within the tropical context (Hii et al., 2011).

In equatorial Singapore (1°21'07.6"N 103°49'11.3"E), the sun's path oscillates bi-annually toward the north- and southward direction, crossing the equinox in March and September, respectively. The higher DLI readings along façades unobstructed from the south can be attributed to direct insolation as a result of the sun's path during the survey period (October–December). This parallels the higher diurnal variability in instantaneous PAR levels observed along façades that experience direct insolation (Fig. 4). Between March–September, it is expected that higher DLI readings and diurnal PAR variability would be observed along façades unobstructed from the north instead. Such large annual variability in the DLI value will be an important factor for designating zones for vertical farming along building façades. Indeed, a comparison across numerous building typologies in the tropical context by Hii et al. (2011) found that throughout the year, the predominantly east–west position of the sun belt near the equator resulted in highest annual insolation along east- and west-facing façades, with the ratio of total insolation between east–west to north–south façades as much as 4–1. Thus, east–west façades are likely to have lower annual PAR variability and higher total annual PAR. Generally, PAR conditions at higher storeys along east–west façades will thus better support the continuous vegetable cultivation in the tropics, especially for basic building typologies with minimal self-shading.

Between façades facing east and west, the largely similar DLI values may be attributed to similar durations of exposure to direct insolation. Owing to the sun's east–west daily sun path, the diurnal PAR distribution pattern shows that east-facing façades experience high PAR levels before the solar noon, while west-facing façades experience high PAR levels after the solar noon (Fig. 4). However, it should be noted that the yearly average DLI value at west-facing façades may be lower, owing to a higher probability of afternoon cloud cover and rainfall in Singapore's tropical climate where convectional rain often occurs in the afternoons (Meteorological Service Singapore, 2015). Despite the lower average DLI value, west-facing façades also experience higher temperatures owing to higher solar heat gain and westward insolation during afternoon hours (Hii et al., 2011).

4.3. Matching crop light requirements

The DLI values along most façades were well below the maximum DLI values of all crops. Though the DLI value along intermediate build-

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ing heights of the façade *S* slightly exceeded the maximum DLI value of lettuce (Fig. 3), PAR conditions would be lower on average owing to weather variability. Based on mean target DLI values, very high-light crops such as kailan, pak choy, and bayam may be grown at façades that experience full-day direct insolation (south or north-facing, depending on the time-of-year), while moderate to high-light crops such as cai xin, kang kong, Chinese cabbage and lettuce may be grown at east and west-facing façades that experience half-day direct insolation. Owing to the lack of direct insolation, façades with low to very low light levels may not be suitable for growth of medium to very high-light vegetable crops.

In addition to the DLI, the assessment of the diurnal PAR distribution is important to ensure that instantaneous PAR values are within the range of crop light requirements. Assuming similar DLI values and a constant light intensity, PAR availability during different times-of-day has shown to have a negligible effect on plant growth (Hughes and Cockshull, 1971). In Bell and Danneberger (1999), no significant differences were found between the growth of plants subjected to morning or afternoon shade. However, analysis of the diurnal distribution showed that PAR levels were not always within the range of plant light requirements. As PAR during mid-day hours at façades that experienced direct insolation exceeded the LSP of medium- to high-light crops, excessive irradiance within several hours of solar noon may result in plant photoinhibition.

In Tan et al. (2014), the suboptimal performance of shade plants in shaded urban conditions were attributed their exposure to sharply contrasting periods of low and high PAR levels characteristic of the built environment. Furthermore, a combination of high temperatures and humidity may result in partial closure of stomata in the afternoon, resulting in reduced photosynthesis (Schulze and Hall, 1982). As photoinhibition may be triggered at higher PAR levels, crops with lower LSP (i.e., lettuce, Chinese cabbage, kang kong) may not be able to withstand extreme conditions as well as crops with higher LSP (i.e., kalian, pak choy, bayam). Thus, even though façades that experience direct insolation have sufficient PAR for the growth of all seven crops, very-high-light crops such as kailan, pak choy, and bayam may be more resilient and suited for growth under the fluctuating light conditions of the built environment, especially at façades that experience direct insolation throughout the day. In order to address excessive PAR levels along building façades, vertical farm designs could incorporate frosted glass, or mechanisms to evenly distribute insolation across different cardinal directions and heights, as in the case of the Sky Greens vertical farm in Singapore (www.skygreens.com).

In the field of architecture and urban design, many simulation models utilise geometric variables to predict solar irradiation for the optimisation of building forms for shading and thermal comfort (Kämpf et al., 2010; Kämpf and Robinson, 2010). Indeed, one of the main goals for urban greenery in building design is to improve the thermal performance of buildings, in order to reduce energy needs for indoor cooling (Hunter et al., 2014). Alongside other conventional passive shading strategies (Valladares-Rendón and Lo, 2014), vertical farming can also be considered a method to delay the transfer of heat into buildings. Further studies could thus explore the optimisation of building forms for both thermal comfort and PAR availability to accommodate the high-light requirements of crop plants.

5. Conclusion

This study highlights the potential for vertical spaces to support urban agriculture, particularly in relation to space-limited conditions within compact cities. Using tropical Singapore as a case-study location, a survey of PAR conditions along the façades of public housing apartment buildings with basic linear and L-shaped configurations show that PAR increased gradually with building height, but remains highly influenced by façade orientation and building forms. Owing to the annual north–south oscillation of the sun path in the tropics, reduced annual PAR variability and higher total annual PAR at façades with an east–west orientation will better support continuous vegetable cultivation, especially for basic building typologies such as the point block configuration. Even though the measured DLI values along façades facing east and west were largely similar, west-facing façades are expected to experience higher average temperatures and a lower average DLI in the tropical context.

Based on leaf physiological traits, the categorisation of the target DLI requirements for seven common leafy vegetables are as follows: kailan, pak choy, and bayam require very high amounts of light (above 30 mol m^{-2} d^{-1}); cai xin requires high amounts of light (20–30 mol m^{-2} d^{-1}); and kang kong, Chinese cabbage, and lettuce require moderate amounts of light (10–20 mol m^{-2} d^{-1}). The results suggest that building façades that experience a minimum of half-day direct insolation will support the growth of vegetables that mostly fall within the moderate to very high-light DLI categories. Notably, highly fluctuating instantaneous PAR conditions may result in plant photoinhibition. Thus, crops with higher light saturation levels such as kailan, pak choy, and bayam may be more resilient to harsh conditions within the built environment. Building-integrated agricultural systems should seek to mitigate the effect of excessive PAR and temperatures during mid-day hours, as well as annual PAR variability that results from the sun's north–south oscillation.

This study adds to the body of knowledge on the spatio-temporal characteristics of PAR along building façades in a compact tropical city, based on a case study of the high-rise and high-density residential environment in Singapore. By highlighting the importance of sufficient PAR alongside conventional optimisation goals in urban design, this study hopes to further inform decisions on design and choice of plants for vertical farming. Together with the development of farming systems that increase crop productivity manifold per unit of land, both micro-gardening and building-integrated agriculture can play a part in achieving greater food self-reliance in space-limited cities.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at https://doi.org/10.1016/j.ufug.2017.11.004.

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